

Business Requirements Specification

The complete, authoritative statement of what the Rozine platform must do — across the Investor app, Business app, Auditor app, Admin console, Pulse and the shared transactional core.

DOCUMENT CONTROL	VALUE
Document	Rozine Platform — Business Requirements Specification
Version	1.0
Status	Baselined for build
Owner	Rozine Technologies Ltd — Product
Audience	Engineering, Compliance, Audit Partner Operations, Regulator, Investors
Supersedes	All prior specifications referencing RRA/EBM tax-data verification (deprecated)
Classification	Confidential

MATERIAL CHANGE FROM PRIOR BASELINE

The verification model has changed fundamentally. Tax-authority integration (RRA sync, EBM device/TIN linkage, automated tax-receipt scraping, tax compliance badging) is **deprecated in full** and must not appear in any interface, data model, contract or communication. It is replaced by bank/MoMo statement ingestion with OCR parsing, and on-site co-signature by ICPAR-accredited Audit Partners under ISRS 4400 agreed-upon procedures.

Purpose, scope and approach

1.1 Purpose

This document specifies, without ambiguity, the business requirements the Rozine platform must satisfy. It is the single reference against which the build is scoped, the compliance programme is evidenced, and delivery is accepted. Where any other artefact conflicts with this document, this document governs.

1.2 In scope

- All five applications: Investor, Business, Auditor, Admin console, Pulse.
- The shared transactional core: entity model, lifecycle state machines, ledger, policy engine, event propagation.
- Underwriting, pricing, rating, verification, monthly reporting, servicing, collections and recovery.
- Primary issuance and secondary-market trading.
- Wallet, payments, treasury and fee collection.
- KYC/KYB, AML, disclosure, governance and supervisory reporting.
- The Audit Partner lifecycle from accreditation to compensation.

1.3 Out of scope

- Any tax-authority integration or tax-derived verification (deprecated — see change notice above).
- Deposit-taking, principal lending from Rozine's balance sheet, or guarantee of investor capital.
- Foreign-currency instruments and cross-border settlement in v1.
- Equity instruments, convertible instruments and revenue-share instruments other than the fixed-return note.
- Insurance products and credit guarantees (may be partnered, not built).

1.4 Requirement notation

PREFIX	CLASS	MEANING
B0-n	Business objective	The outcome the platform exists to produce.
BR-n	Business rule	An invariant the system must never violate.
FR-n	Functional requirement	A capability the system must provide.
DR-n	Data requirement	Data that must be captured, derived or retained.
NFR-n	Non-functional requirement	A quality attribute or operational constraint.
IR-n	Integration requirement	An external system dependency.
CR-n	Compliance requirement	A regulatory or legal obligation.
AC-n	Acceptance criterion	The condition under which delivery is accepted.

MUST denotes a mandatory requirement. **SHOULD** denotes a strong recommendation that may be deferred with documented justification. **MAY** denotes an option.

Business objectives

ID	OBJECTIVE	MEASURE OF SUCCESS
B0-1	Enable profitable, asset-light Rwandan businesses to raise growth capital without collateral.	Capital deployed; number of distinct businesses funded; median days from application to funded.
B0-2	Give retail investors a comprehensible, verified, fixed-return instrument accessible from RWF 5,000.	Registered investors; median first-ticket size; reinvestment rate at maturity.
B0-3	Make monthly verification of small-business performance cheap, rigorous and routine.	Cost per verified report; report SLA compliance rate; parsing accuracy against auditor ground truth.
B0-4	Never permit a business to borrow beyond its verified debt-service capacity.	Zero notes originated above computed capacity; portfolio at risk by cohort.
B0-5	Provide pre-maturity liquidity to investors through a functioning secondary market.	Secondary volume; median time-to-fill; spread to par.
B0-6	Operate to a supervisory standard from day one, sufficient to enter and exit the RCMA sandbox successfully.	Sandbox admission; zero material findings; licence granted.
B0-7	Earn only from platform services, never from spread, distress or principal risk.	Revenue composition across the five fee lines; zero revenue derived from default.
B0-8	Build a durable, disciplinable Audit Partner profession with aligned economics.	Accredited partners; partner retention; yield-share paid; disciplinary incidence.

Actors, roles and permissions

3.1 Primary actors

ACTOR	APP	RESPONSIBILITY
Investor	Investor	Funds notes, holds positions, receives repayments and verified monthly reports, trades on the secondary market.
Business owner	Business	Applies, uploads statements, accepts capacity, creates and manages a raise, submits monthly reports, repays.
Audit Partner	Auditor	ICPAR-accredited CPA. Performs on-site verification, captures evidence, co-signs reports, earns yield share.
Rozine staff	Admin	Underwriting, compliance, treasury, dispatch, partner accreditation, policy, dispute resolution.
Prospect	Pulse	Pre-launch investor pledging or business pre-qualifying. Holds no account and transacts no funds.
Supervisor	Admin (read-only)	Regulator or auditor granted a read-only seat over the live operating state.

3.2 Staff roles and permission matrix

CAPABILITY	ANALYST	APPROVER	TREASURY	COMPLIANCE	SUPERADMIN
View all directories and records	Yes	Yes	Yes	Yes	Yes
Approve / decline an application	No	Yes	No	No	Yes
Override computed capacity	No	No	No	No	Yes
Verify ICPAR licence, accredit partner	No	Yes	No	Yes	Yes
Freeze / unfreeze a party	No	No	No	Yes	Yes
Freeze / release Audit Partner yield	No	Yes	Yes	Yes	Yes
Post manual ledger entry / reversal	No	No	Yes	No	Yes
Halt or resume the secondary market	No	No	Yes	Yes	Yes
Amend underwriting or fee policy	No	No	No	No	Yes
Toggle app feature flags / maintenance	No	No	No	No	Yes
Manage staff accounts and roles	No	No	No	No	Yes
Act-as (view a user's live app state)	Yes	Yes	No	Yes	Yes
Broadcast to users	No	Yes	No	Yes	Yes

ID	REQUIREMENT
BR-1	Every administrative action MUST be gated by the actor's role. An ungated action is a defect.
BR-2	Every administrative action MUST write an immutable audit entry recording actor, target, action, stated reason and timestamp. Reason is mandatory and non-empty.
BR-3	Destructive or financially material actions (freeze, reversal, capacity override, yield release, market halt, policy amendment) MUST require explicit confirmation, and MUST support dual approval where policy requires it.
BR-4	Act-as MUST be read-only, visibly indicated, time-boxed, and logged as a distinct event class.
BR-5	No role, including Superadmin, MAY delete an audit entry or a ledger entry.

Domain model and data requirements

4.1 Core entities

ENTITY	KEY ATTRIBUTES
Party	Party ID, kind (business / investor / auditor / staff / platform), legal name, KYC-KYB state, verification tier, frozen flag, contact, district and province, created-at.
Business profile	Registration number, sector, premises geolocation, owner identity, bank and MoMo accounts on file, statement history, derived cash-flow series, capacity, rating, health status, standing.
Auditor profile	ICPAR member ID, practising certificate number, licence expiry, accreditation state, office geolocation, active engagement count and cap, performance rating, yield-frozen flag.
Application	Application ID, business, requested tenor, computed capacity, DSCR, tier, decision, decision reason, underwriter, timestamps.
Note	Note ID, issuer, title, state, target amount, raised amount, tenor months, total return rate, rating at issue, unit price, units outstanding, listing date, expiry date, maturity date, assigned Audit Partner.
Holding	Investor, note, units, cost basis, accrued return, payouts received, acquisition channel (primary / secondary).
Monthly report	Report ID, note, business, Audit Partner, reporting period, state, statement document reference, parsed gross inflow and outflow, liquidity coverage, health status, business note and photographs, auditor note and geo-tagged photographs, verification seal hash, submission and co-signature timestamps.
Order	Order ID, note, seller, units, ask price, state, matched buyer, settlement reference.
Ledger entry	Entry ID, kind, debit party, credit party, amount, currency, reference, reversal-of, immutable timestamp.
Repayment schedule	Note, instalment number, due date, principal component, return component, fee components, state, paid-at.
Dispute	Dispute ID, report, raising party, respondent, reason, state, resolving staff, resolution note.
Policy	Path, value, version, amended-by, amended-at, prior value.
Audit action	Action ID, staff actor, target type and ID, action performed, reason note, timestamp.
Notification	Recipient, class, subject, body, related entity, read state, delivery channels.
Pulse registration	Side (investor / business), name, contact method and value, province and district, pledge amount or pre-qualified amount, sequence number, created-at.

4.2 Data requirements

ID	REQUIREMENT
DR-1	All monetary amounts MUST be stored as exact decimal values in Rwandan francs. Floating-point representation of money is prohibited.
DR-2	Every balance MUST be derived from immutable ledger entries. No stored balance field MAY be mutated directly.
DR-3	Every uploaded statement MUST be retained in original form alongside its parsed derivation, with both linked to the report or application that used them.
DR-4	Every evidence photograph MUST retain capture timestamp, device-reported geolocation and capture method (in-app camera only).
DR-5	Every underwriting decision MUST retain its complete input set, so the decision can be recomputed and explained years later.
DR-6	Policy values MUST be versioned. A decision MUST record which policy version governed it.
DR-7	Personal data MUST be classified, access-controlled and retained per the retention schedule; deletion requests MUST be honoured except where records are required by law or regulation.
DR-8	The system MUST NOT store any tax identification number, EBM device identifier, or tax-authority credential. Existing columns MUST be dropped.

Lifecycles

5.1 Note lifecycle

STATE	ENTERED WHEN	PERMITTED TRANSITIONS
Draft	Business begins a raise.	→ Submitted, → Withdrawn
Submitted	Business submits for underwriting.	→ Audit, → Declined
Audit	Underwriting cleared; awaiting Audit Partner co-signature.	→ Approved, → Declined
Approved	Co-signed and cleared for listing.	→ Live
Live	Published and accepting investment.	→ Funded, → Expired, → Withdrawn
Funded	Target reached; disbursement executed.	→ Repaying
Repaying	Schedule active.	→ Matured, → Arrears
Arrears	A scheduled instalment is unpaid past grace.	→ Repaying, → Default
Matured	All instalments settled.	Terminal
Default	Recovery process initiated.	→ Recovered, → Written off
Expired	Listing validity elapsed without full funding.	Terminal; committed funds returned

5.2 Monthly report lifecycle

STATE	DESCRIPTION
Drafting	Reporting window open; business preparing upload.
Submitted	Business has uploaded the statement, note and photographs. Dispatch triggered.
Pending audit	Assigned to an Audit Partner; on-site visit outstanding.
Co-signed	Partner has completed procedures, attached evidence and applied the verification seal.
Published	Visible to all holders of the note. Immutable thereafter.
Disputed	A party has raised a dispute; under Admin resolution.
SLA breached	Window closed without publication. Escalation and yield freeze triggered.

5.3 Audit Partner lifecycle

Pending verification → ICPAR licence checked → engagement terms signed in-app → biometrics bound → Active → (Suspended / Yield-frozen) → Reinstated or Terminated. Licence expiry MUST automatically suspend dispatch eligibility.

Business rules

6.1 Underwriting and capacity

ID	RULE
BR-10	Capacity MUST be computed from verified cash flow. A business MUST NOT be able to request an amount; it accepts an offer at or below computed capacity.
BR-11	A note MUST NOT be listed for more than the issuer's computed capacity at the time of listing.
BR-12	Capacity MUST account for existing Rozine obligations and any disclosed external debt service.
BR-13	Capacity computation MUST use a volatility-trimmed measure of net operating cash flow so that a single exceptional month cannot inflate the offer.
BR-14	DSCR below 1.00× MUST result in decline. DSCR between 1.00× and 1.25× MUST require enhanced procedures and manual approval. DSCR at or above 1.25× MAY auto-approve subject to co-signature and KYB.
BR-15	A minimum verified statement history MUST be present before any capacity is offered.
BR-16	A declined business MUST retain its profile, continue to be re-scored as new statements arrive, and be re-eligible automatically without reapplying from zero.
BR-17	Capacity override MUST be restricted to Superadmin, require a stated reason, and be logged and surfaced on the note record permanently.

6.2 Pricing and terms

ID	RULE
BR-20	Permitted tenors are exactly 3, 6, 9 and 12 months. No other tenor MAY be offered.
BR-21	Return MUST be expressed as a flat total return on principal , not as an annualised rate. The term "APR" MUST NOT appear in any user-facing surface.
BR-22	Total return MUST be at least 10% and at most 15% of principal, on every note, at every tenor, without exception.
BR-23	Total return MUST be a function of rating and tenor within those bounds, and MUST be deterministic and explainable.
BR-24	Repayment MUST be in fixed monthly instalments comprising principal and return, known in full at the time of investment.
BR-25	Early repayment MUST NOT reduce the total return owed to investors.
BR-26	The business MUST see the total cost of capital in francs, and the investor MUST see the total amount returned in francs, before committing.
BR-27	Minimum investment is RWF 5,000. Units MUST be priced so that this minimum is always achievable.
BR-28	A listing MUST expire 30 days after going live if not fully funded; committed funds MUST be returned in full with no fee.

6.3 Rating

ID	RULE
BR-30	The platform MUST publish exactly one measure of business quality. Any second competing score is prohibited.
BR-31	The rating MUST be expressed on a 0.0–5.0 scale with one decimal place, everywhere, in every app including Admin and Auditor.
BR-32	Bands MUST be: Strong ≥ 4.0 ; Stable ≥ 3.0 ; Weak ≥ 2.0 ; Distressed < 2.0 . Band boundaries MUST fall on whole numbers.
BR-33	Presentation MUST lead with the band word, support it with the numeric rating, and reinforce with the band colour.
BR-34	"Stable" MUST be presented as the healthy default, never as a warning state.
BR-35	A business rated Distressed MUST NOT be listable. Its existing notes enter active monitoring.
BR-36	The rating MUST be recomputed on every verified monthly report and on any material event.
BR-37	The rating at time of issue MUST be recorded immutably on the note; the live rating MUST also be shown to holders.

6.4 Verification and reporting

ID	RULE
BR-40	No note MAY be listed without an Audit Partner co-signature on the originating verification.
BR-41	The monthly reporting window is the 1st to the 7th inclusive. Submission and co-signature MUST both occur within it.
BR-42	A report MUST NOT be published without a valid verification seal from an accredited, unexpired Audit Partner.
BR-43	Evidence photographs MUST be captured through the in-app camera with live geolocation. Gallery upload MUST be rejected.
BR-44	Captured geolocation MUST be cross-checked against the registered premises; a material discrepancy MUST flag the report for review.
BR-45	A published report MUST be immutable. Corrections MUST be issued as an amendment linked to the original, never as an edit.
BR-46	An SLA breach MUST freeze the responsible Audit Partner's yield share pending Admin resolution, and MUST flag the note to holders.
BR-47	The business note and the auditor note are each limited to 100 characters; photographs are limited to five per party per report.
BR-48	Flash Audits MUST be randomly triggerable with a 24-hour completion requirement, and MUST be indistinguishable in advance from routine dispatch.

6.5 Audit Partner integrity

ID	RULE
BR-50	Only a verified ICPAR member with a current practising certificate MAY be accredited or receive dispatch.
BR-51	Dispatch MUST be restricted to partners whose office is within 30 km of the business premises, subject to capacity.
BR-52	A partner MUST NOT exceed the configured concurrent-engagement cap.
BR-53	Assignments MUST rotate; a partner MUST NOT steward the same business indefinitely.
BR-54	A partner MUST NOT be assigned to a business with which a declared conflict of interest exists.
BR-55	Partner compensation is 25% of the service fee collected on notes they steward, payable monthly on satisfactory SLA performance.
BR-56	Licence expiry MUST automatically suspend dispatch eligibility without staff action.

6.6 Money and fees

ID	RULE
BR-60	Investor funds MUST be held in segregated accounts and MUST NOT be commingled with Rozine operating funds at any time.
BR-61	Rozine MUST NOT take principal risk, guarantee capital, or represent any note as capital-protected.
BR-62	The fee schedule is: listing fee (business, fixed, on listing); service fee 2% of each repayment (business); repayment fee 1% of each payout (investor); secondary fee 3% of each trade (seller). No other fee MAY be charged.
BR-63	25% of every service fee collected MUST be routed to the stewarding Audit Partner.
BR-64	All fees MUST be disclosed in francs before the user commits to the action that incurs them.
BR-65	Every movement of value MUST produce balanced double-entry ledger records. Reversals MUST be posted as new compensating entries referencing the original.
BR-66	Disbursement MUST occur only after the note is fully funded and all conditions precedent are satisfied.
BR-67	Repayments MUST be allocated to holders pro rata to units held on the record date.

6.7 Secondary market

ID	RULE
BR-70	Only fully settled holdings in notes that are Repaying and current MAY be listed for sale.
BR-71	Holdings in notes in Arrears, Default or Disputed status MUST NOT be tradable.
BR-72	The buyer MUST see the note's current rating, health status, latest published report and remaining schedule before purchase.
BR-73	Settlement MUST be atomic: units and funds transfer together or not at all.
BR-74	Admin MUST be able to halt the secondary market globally or per note, immediately.
BR-75	Rozine MUST NOT act as principal, market maker or price-setter on the secondary market.

6.8 Pulse

ID	RULE
BR-80	Pulse MUST NOT collect funds, issue any instrument, or create any binding obligation on either party.
BR-81	Every Pulse surface MUST carry a clear non-binding demand-simulation disclosure.
BR-82	A pre-qualification figure MUST be computed by the same engine used in production, and MUST be labelled indicative and subject to full verification.
BR-83	Registrations MUST capture name, one contact method (phone or email), province and district, and MUST issue a sequence number.
BR-84	Pulse counters displayed publicly MUST reflect real registration activity, never fabricated figures.

Functional requirements — Investor app

ID	CAPABILITY	REQUIREMENT
FR-100	Onboarding	Register, verify identity to the required KYC tier, accept terms and risk disclosure, and set up a wallet before any investment is permitted.
FR-101	Deal discovery	Browse live notes as a card deck with filters by sector, tenor, rating band and yield, and a curated ordering.
FR-102	Deal detail	View verified financials, rating and its basis, funding progress, tenor, total return, repayment schedule, use of funds, business photographs, and the Audit Partner's identity and licence.
FR-103	Evidence access	Open every published monthly report for the note, including parsed figures, both parties' notes and photographs, geolocation and the verification seal.
FR-104	Investment	Select units, see the exact amount invested and amount returned in francs, review all fees, and confirm. Minimum RWF 5,000.
FR-105	Watchlist & follow	Save notes and follow businesses to be alerted to new raises.
FR-106	Portfolio	View total value, invested principal, projected return, positions, upcoming payouts, diversification by sector, rating mix, and idle wallet balance.
FR-107	Holding detail	Per position: schedule, payouts received and outstanding, current rating and health, arrears warnings, and the report history.
FR-108	Secondary market	List a holding for sale at a chosen price, browse and purchase others' listings, and track order state to settlement.
FR-109	Wallet	Deposit, withdraw, view balance and a complete transaction history with per-transaction detail.
FR-110	Notifications	Receive and review alerts for new reports, payouts, funding milestones, arrears, disputes and platform notices.
FR-111	Education	Access lessons and FAQs explaining the rating, the fixed-return structure, fees, risk and the secondary market.
FR-112	Profile	Manage personal information, KYC documents, bank and MoMo details, security settings, notification preferences, statements and support.
FR-113	Risk disclosure	Present unavoidable, plain-language risk disclosure at first investment and on every deal page. Capital loss MUST be stated explicitly.

Functional requirements — Business app

ID	CAPABILITY	REQUIREMENT
FR-200	Onboarding & KYB	Register the business, capture registration details, owner identity, premises geolocation and contact; complete KYB to the required tier. No tax identifier MAY be requested.
FR-201	Statement ingest	Upload bank and MoMo statements as PDF. Accepted evidence types: MoMo statement, bank statement, audited cash transactions, POS data.
FR-202	Parsing feedback	Show parsed inflow, outflow and derived series for confirmation, with the ability to flag a parsing error for review.
FR-203	Capacity & rating	Display computed capacity, rating with band, indicative total return by tenor, and a plain-language explanation of each.
FR-204	Create a raise	Choose an amount at or below capacity, choose a tenor from 3/6/9/12, state use of funds, attach photographs, and preview the exact total cost in francs and the monthly repayment.
FR-205	Submission & tracking	Submit for underwriting and track state through audit, approval, listing, funding and disbursement, with reasons shown at every step.
FR-206	Monthly report	Within the 1st–7th window: upload the period statement, add a note (≤100 characters) and up to five photographs, and submit for Audit Partner co-signature.
FR-207	Audit coordination	See the assigned Audit Partner, expected visit, co-signature state, and any outstanding request from the partner.
FR-208	Repayments	View the full schedule, pay an instalment, see fees applied, and receive arrears warnings before and after a due date.
FR-209	Standing & health	See current rating, health status, reporting compliance history and the specific factors improving or damaging standing.
FR-210	Dispute	Raise a dispute against a report or an audit finding, with reason and evidence, and track resolution.
FR-211	Wallet	Receive disbursement, hold balance, withdraw to a registered bank or MoMo account, and view full transaction history.
FR-212	Education	Access guidance on improving rating, managing cash flow, preparing for an audit visit, and understanding fees.
FR-213	Decline handling	On decline, present the computed figures, the specific shortfall, and the concrete conditions under which the business would qualify.

Functional requirements — Auditor app

ID	CAPABILITY	REQUIREMENT
FR-300	Accreditation	Register, submit ICPAR member ID and practising certificate, complete biometric binding, and await verification. Dispatch MUST be blocked until Active.
FR-301	Engagement signing	Review and execute the master services agreement and ISRS 4400 agreed-upon-procedures terms inside the app, with a retained signed record.
FR-302	Job queue	Receive dispatched jobs within the 30 km radius, with business detail, location, deadline and job type (routine or Flash Audit).
FR-303	Flash Audit timer	Display a live 24-hour countdown for Flash Audits with escalating urgency and breach consequence stated.
FR-304	On-site capture	Capture geo-tagged photographs through the in-app camera only. Gallery selection MUST be unavailable.
FR-305	Procedures checklist	Work through the agreed-upon procedures as a structured checklist, recording findings against each.
FR-306	Statement verification	Review the parsed figures against source documents and on-site observation; confirm, correct or reject with reason.
FR-307	Co-signature	Apply the verification seal, binding auditor identity, licence, report contents and timestamp. Seal MUST be irreversible.
FR-308	Portfolio	View all stewarded businesses, their reporting history, health and upcoming obligations.
FR-309	Earnings	See yield share accrued, paid and frozen, per note and in total, with the basis of each calculation.
FR-310	Academy	Complete continuing-education modules covering the platform's metrics, formulas, procedures and ethics, with completion recorded.
FR-311	Offline tolerance	Complete a full field capture without connectivity and synchronise reliably when connectivity returns, without data loss.
FR-312	Conflict declaration	Declare a conflict of interest and decline a job, with the declaration recorded and the job re-dispatched.

SECTION 10

Functional requirements — Admin console

The Admin console **MUST** provide complete visibility and command over every entity and action in the ecosystem. Any operation possible in the system **MUST** be observable and, where appropriate, reversible from Admin.

ID	AREA	REQUIREMENT
FR-400	Command dashboard	Present live platform state: capital deployed, active notes, funding health, attention items requiring action, and quick commands.
FR-401	Business directory	Search, filter and inspect every business; view rating, capacity, notes, reports, standing and full history; freeze, adjust KYB tier, or override capacity.
FR-402	Investor directory	Search, filter and inspect every investor; view KYC state, wallet, holdings, orders and activity; freeze or adjust tier.
FR-403	Auditor network	ICPAR verification desk for pending accreditations with licence detail and manual override; spatial dispatch map showing partner radius, capacity and load; reassign engagements; suspend, terminate or reinstate; freeze or release yield.
FR-404	Application queue	Review submitted applications with full underwriting basis; approve, decline or return with reason; enforce tier rules.
FR-405	Note management	Inspect any note across its entire lifecycle; force state transitions where legitimate, with reason and log.
FR-406	Monthly audit compliance	Board of every report by state (Drafting, Pending audit, Published, SLA breached, Disputed); open any report in a detail panel with full evidence; escalate, reassign, freeze yield or resolve.
FR-407	Secondary market	Monitor orders and settlement; halt or resume trading globally or per note; cancel an order with reason.
FR-408	Treasury & ledger	View the full double-entry ledger; reconcile segregated accounts; post manual entries and reversals under dual approval; monitor fee revenue by line.
FR-409	Risk & distress	Distressed-note board with outstanding amount, days late, risk band and probability of default; open a review panel with the why-flagged basis, recommended action and one-click execution.
FR-410	Compliance	Case management for AML alerts and investigations; document review; freeze powers; suspicious-activity recording.
FR-411	Policy engine	Amend every configurable parameter — DSCR thresholds, tenor set, return floor and ceiling, fee rates, yield share, dispatch radius, concurrency caps, SLA windows, exposure limits — with versioning and attribution.
FR-412	App control	Per-app feature flags, maintenance mode and forced-update controls for Investor, Business, Auditor and Pulse.
FR-413	Staff & roles	Create, modify and disable staff accounts; assign roles; review each role's permission set.
FR-414	Audit trail	Complete, immutable, searchable record of every administrative action with actor, target, action, reason and timestamp.
FR-415	Act-as	View any user's live application state read-only, clearly indicated and logged.
FR-416	Messaging	Broadcast to a targeted audience across apps; manage templates and automated notification rules.
FR-417	Disputes	Intake, investigate and resolve disputes between businesses and Audit Partners, with the outcome recorded on the report.

ID	AREA	REQUIREMENT
FR-418	Supervisory reporting	Generate and export the reports required by the regulator, and support a read-only supervisor seat.

Functional requirements — Pulse and shared core

11.1 Pulse

ID	CAPABILITY	REQUIREMENT
FR-500	Investor pledge	Set an intended amount, see the projected total return, register name, contact method, province and district, and receive a sequence number.
FR-501	Business pre-qualification	Upload one statement and receive a real computed pre-qualified amount, rating and indicative yield, produced by the production engine.
FR-502	Shareable pass	Issue a downloadable and shareable pass carrying the registrant's name, figure and sequence number, for organic distribution.
FR-503	Live counters	Display real, live aggregates of registration activity on both sides.
FR-504	Sample deals	Show a rotating sample of pre-qualified businesses with term, yield, rating and the amount a pledger would earn.
FR-505	Disclosure	State prominently that this is a non-binding demand simulation and that no funds are collected or issued.

11.2 Shared transactional core

ID	CAPABILITY	REQUIREMENT
FR-600	Single source of truth	All apps MUST read and write the same core. No app MAY hold private authoritative state.
FR-601	Live propagation	A committed change MUST be reflected across every affected app without a manual refresh.
FR-602	Deterministic computation	Capacity, DSCR, rating, pricing, schedules, dispatch and fee calculations MUST be reproducible from stored inputs.
FR-603	Idempotency	Every financial operation MUST be idempotent; a retried request MUST NOT double-post.
FR-604	Event log	Every state transition MUST emit a durable event usable for notification, analytics and reconstruction.
FR-605	Policy as data	All thresholds and rates MUST be configurable data, versioned and attributed — never hard-coded.
FR-606	Expiry engine	Automatically expire listings at 30 days, return committed funds, and notify all parties.
FR-607	Scheduler	Open and close the monthly reporting window, trigger dispatch, escalate SLA breaches, release or freeze yield, and drive repayment due dates.

Computation specification

NORMATIVE FORMULAS

$$\text{NOCF}_m = \text{Inflow}_m - \text{Outflow}_m$$
$$\text{CFADS} = \text{trimmed_mean}(\text{NOCF}) - \text{ExistingDebtService} - \text{OwnerDraw}$$
$$\text{DSCR} = \text{CFADS} \div \text{ProposedMonthlyRepayment}$$
$$\text{Capacity} = (\text{CFADS} \div \text{TargetDSCR} \times \text{Tenor}) \div (1 + \text{TotalReturn})$$
$$\text{Instalment} = (\text{Principal} \times (1 + \text{TotalReturn})) \div \text{Tenor}$$
$$\text{Rating} = \text{clamp}(\text{EngineScore} \div 20, 0.0, 5.0) \text{ rounded to 1 dp}$$
$$\text{Coverage} = \text{Inflow}_{\text{period}} \div (\text{Outflow}_{\text{period}} + \text{DebtService}_{\text{period}})$$

Figure 1 — All monetary results round to the nearest franc; rates round to one decimal place. Capacity is solved from serviceable repayment, never from a requested amount.

ID	REQUIREMENT
FR-700	Rounding MUST be applied consistently and specified per computation; the sum of instalments MUST exactly equal principal plus total return, with any residual applied to the final instalment.
FR-701	Every computed figure shown to a user MUST be traceable to its inputs and reproducible on demand by Admin.
FR-702	Where a computation depends on a policy value, the governing policy version MUST be recorded with the result.

Non-functional requirements

ID	ATTRIBUTE	REQUIREMENT
NFR-1	Availability	Core transactional services MUST target high availability, with planned maintenance announced in advance and surfaced through app maintenance mode.
NFR-2	Data integrity	No committed financial transaction MAY be lost or double-applied under any failure mode. Reconciliation MUST detect and report any imbalance.
NFR-3	Security	Encryption in transit and at rest; managed key rotation; least-privilege access; secrets never in source; regular penetration testing.
NFR-4	Authentication	Strong authentication for all parties; multi-factor for staff and Audit Partners; biometric binding for Audit Partners; session expiry and revocation.
NFR-5	Auditability	Every state change MUST be attributable to an actor and reconstructible in sequence.
NFR-6	Performance	Interactive screens MUST remain responsive on mid-range Android devices over intermittent mobile networks; statement parsing MAY be asynchronous with clear progress feedback.
NFR-7	Resilience	The Auditor app MUST function fully offline in the field and synchronise without loss.
NFR-8	Accessibility	Adequate contrast, legible minimum type sizes, and touch targets no smaller than 44 px.
NFR-9	Localisation	Architecture MUST support Kinyarwanda, English and French; all user-facing strings externalised.
NFR-10	Clarity	Financial figures MUST be shown in full francs with thousands separators where precision matters. No user-facing jargon without an inline explanation.
NFR-11	Scalability	The platform MUST scale to thousands of concurrent monthly audits without a proportional increase in Rozine operations headcount.
NFR-12	Recoverability	Backups MUST be automated, encrypted, geographically separated and restore-tested on a defined cycle.
NFR-13	Observability	Structured logging, metrics and alerting across all services, with dashboards for the platform KPIs.
NFR-14	Data residency	Personal and financial data MUST be stored in compliance with Rwandan data-protection requirements.

Integrations and compliance

14.1 Integrations

ID	SYSTEM	PURPOSE
IR-1	Mobile money providers	Disbursement, collection, wallet funding and withdrawal.
IR-2	Commercial banks	Segregated account operation, settlement and reconciliation.
IR-3	Identity verification	Individual identity confirmation for KYC.
IR-4	Business registry	Confirmation of registration and ownership for KYB.
IR-5	ICPAR register	Verification of member ID, practising certificate and standing.
IR-6	Messaging providers	SMS, email and push delivery of notifications.
IR-7	Mapping / geocoding	Premises geolocation, dispatch radius and evidence cross-check.
IR-8	Supervisory reporting	Delivery of periodic returns to the regulator.

14.2 Compliance requirements

ID	REQUIREMENT
CR-1	KYC and KYB MUST be completed and verified before any transaction is permitted, with tiered limits by verification level.
CR-2	AML transaction monitoring MUST run continuously, with case management, escalation and regulatory reporting.
CR-3	Sanctions and PEP screening MUST be performed at onboarding and periodically thereafter.
CR-4	Investor funds MUST be segregated and reconciled daily.
CR-5	Every listing MUST carry complete disclosure: verified financials, rating basis, all fees, tenor, schedule and explicit risk warning.
CR-6	The platform MUST NOT represent, imply or permit marketing suggesting capital protection or guaranteed return.
CR-7	Investment limits appropriate to investor category MUST be enforceable by policy.
CR-8	A documented complaints and dispute-resolution process MUST be available to all users and evidenced in the system.
CR-9	Records MUST be retained for the statutory period and be producible on regulatory request.
CR-10	Personal data processing MUST comply with Rwandan data-protection law, with lawful basis, consent capture and subject-rights handling.
CR-11	Audit Partner engagements MUST be performed under ISRS 4400 agreed-upon procedures with retained working papers.
CR-12	The platform MUST support a read-only supervisor seat providing live visibility of the operating state.
CR-13	Sandbox caps and conditions MUST be enforceable as policy values, without code change.

Assumptions, constraints, dependencies

CLASS	STATEMENT
Assumption	Target businesses transact predominantly through bank and mobile-money channels, producing machine-readable statement history.
Assumption	ICPAR-accredited accountants are available in sufficient number and geographic spread to serve the dispatch radius.
Assumption	The RCMA sandbox remains an available pathway for a platform of this type.
Assumption	Retail investors will accept a fixed-return, capital-at-risk instrument when verification is credible and disclosure is plain.
Constraint	Rozine MUST NOT take deposits, lend as principal, or guarantee capital.
Constraint	All instruments are denominated in Rwandan francs in v1.
Constraint	Tenors are limited to 3, 6, 9 and 12 months; total return is bounded at 10–15% flat.
Constraint	No tax-authority data may be used or stored anywhere in the system.
Dependency	Payment-rail agreements with mobile-money providers and a banking partner for segregated accounts.
Dependency	A working relationship with ICPAR for licence verification and disciplinary referral.
Dependency	Regulatory admission to the sandbox before live operation.
Dependency	Parsing accuracy sufficient across the major statement formats in market.

Acceptance criteria

ID	CRITERION
AC-1	A business can be onboarded, verified, statement-parsed, capacity-computed, co-signed, listed, funded, disbursed, serviced across a full schedule, and matured — end to end, on production rails.
AC-2	An investor can register, verify, fund a wallet, invest from RWF 5,000, receive a monthly verified report, receive scheduled payouts, sell on the secondary market, and withdraw.
AC-3	An Audit Partner can be accredited against the ICPAR register, sign engagement terms in-app, receive a dispatched job, complete an on-site capture offline, and apply a verification seal that publishes to holders.
AC-4	Admin can observe and command every entity and action described in Section 10, with every action gated by role and logged with a reason.
AC-5	No note can be created above computed capacity, at a disallowed tenor, or outside the 10–15% return band, by any path including Admin override without explicit Superadmin action and log.
AC-6	A missed reporting window automatically breaches SLA, freezes the partner's yield, flags the note and notifies all affected parties without staff intervention.
AC-7	The ledger reconciles exactly: assets held equal obligations owed, across every account, every day.
AC-8	A regulator with a read-only seat can inspect the live operating state and reconstruct any historical decision from stored inputs.
AC-9	No surface in any app, or any stored record, contains a reference to RRA, EBM, TIN, tax sync or a tax compliance badge.
AC-10	Every app displays the rating identically: one word, one number out of five to one decimal, one colour.
AC-11	Pulse operates with no funds collected and no instrument issued, and its counters reflect real registrations.
AC-12	A penetration test and an independent security review are completed with no unresolved high-severity finding.

Parameters and glossary

A. Configurable parameters (policy-managed)

PARAMETER	BASELINE VALUE
Permitted tenors	3, 6, 9, 12 months
Total return floor / ceiling	10% / 15% flat on principal
Minimum investment	RWF 5,000
Auto-approve DSCR	≥ 1.25×
Audit-band DSCR	1.00× – 1.25×
Rating band thresholds	4.0 / 3.0 / 2.0
Reporting window	1st – 7th of month
Flash Audit completion	24 hours
Dispatch radius	30 km
Audit Partner yield share	25% of service fee
Service fee	2% of each repayment
Repayment fee	1% of each payout
Secondary fee	3% of trade value
Listing validity	30 days
Note / photo limits	100 characters; 5 photographs per party per report

B. Glossary

TERM	DEFINITION
Note	The instrument issued by a business and held by investors, carrying a fixed total return and fixed monthly repayments.
Capacity	The maximum a business may raise, computed from verified cash flow. Never negotiable upward.
CFADS	Cash flow available for debt service — verified operating cash flow after existing obligations and owner drawings, volatility-adjusted.
DSCR	Debt service coverage ratio — CFADS divided by proposed monthly repayment.
Audit Partner	ICPAR-accredited CPA contracted to perform on-site verification and co-sign reports under ISRS 4400.
Verification seal	Cryptographic hash binding auditor identity, licence, report contents and timestamp to a published report.
Flash Audit	Randomly triggered 24-hour re-verification of a business and its Audit Partner.
Health status	Period-level condition derived from the verified report: Healthy, Watch or Distressed.
ISRS 4400	International Standard on Related Services governing agreed-upon-procedures engagements.
ICPAR	Institute of Certified Public Accountants of Rwanda.
RCMA	Rwanda Capital Market Authority.
Pulse	Pre-launch demand-aggregation portal. Collects no funds and issues no instrument.

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